

Infrared Remote Signal Collection

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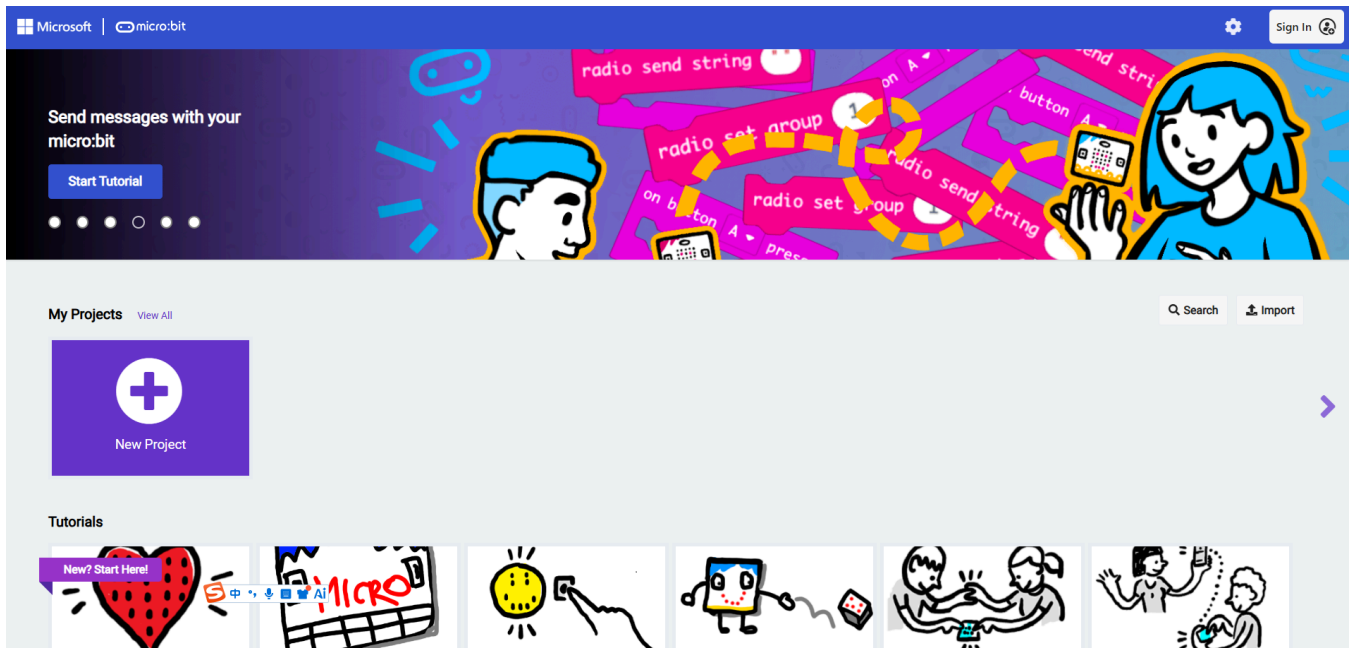
1. Experiment Purpose

The Lastbit car is equipped with an HS0038B infrared receiver module on board, with the silk screen marking IRQ. The HS0038B is a highly integrated **38kHz infrared remote control receiver module**. When the infrared remote control is pressed, the driven infrared LED flashes at high speed (invisible to the human eye). After the infrared receiver probe recognizes the infrared signal, the module's internal circuit processes the signal through optical filtering, demodulation, and shaping before outputting an electrical level signal. It can be used with an infrared remote control (NEC protocol). This lesson will guide you through learning how to read infrared remote control buttons and execute display or control functions according to different buttons.

The standard NEC protocol sends **4 bytes (32 bits)** of data for each key press, with the following structure:

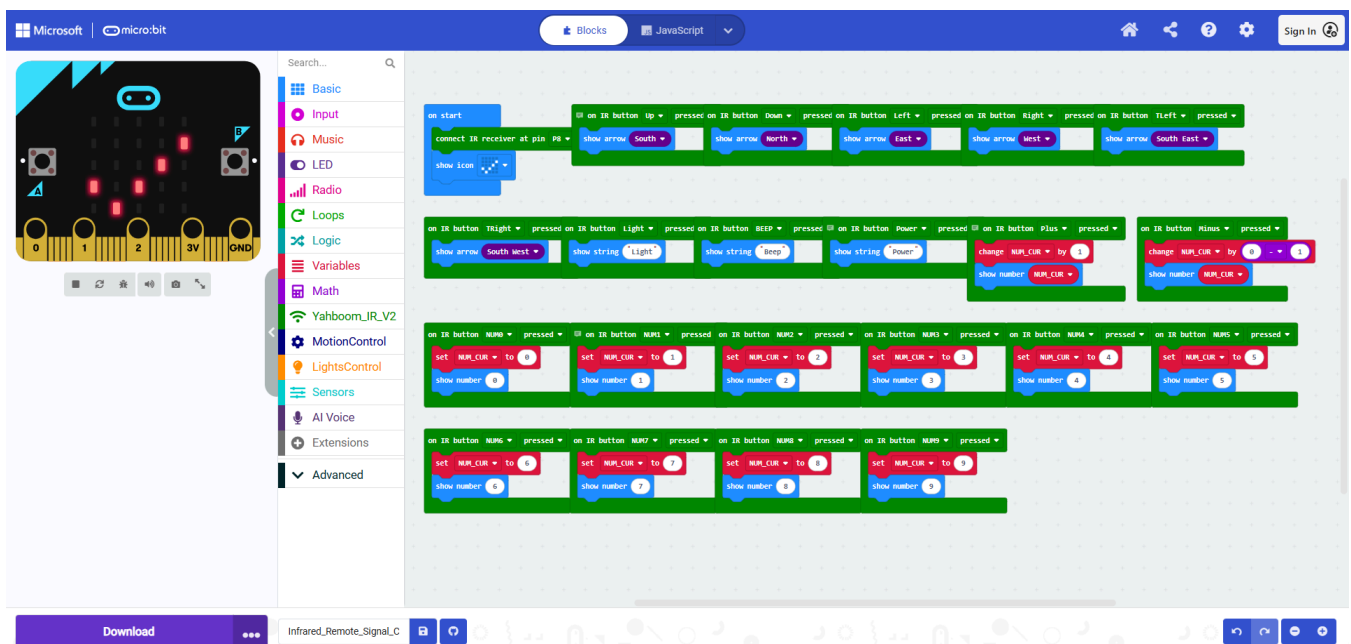
Leader Code → Address Code → Address Inverse Code → Command Code → Command Inverse Code → End Bit

- **Address Code:** Identifies the device type (such as TV, DVD) to prevent different devices from interfering with each other.
- **Address Inverse Code (~Address):** The bitwise inverse of the address code, used to verify that the address transmission is correct.
- **Command Code:** The specific button command (such as volume+, power off).
- **Command Inverse Code (~Command):** The bitwise inverse of the command code, used to verify that the command transmission is correct.

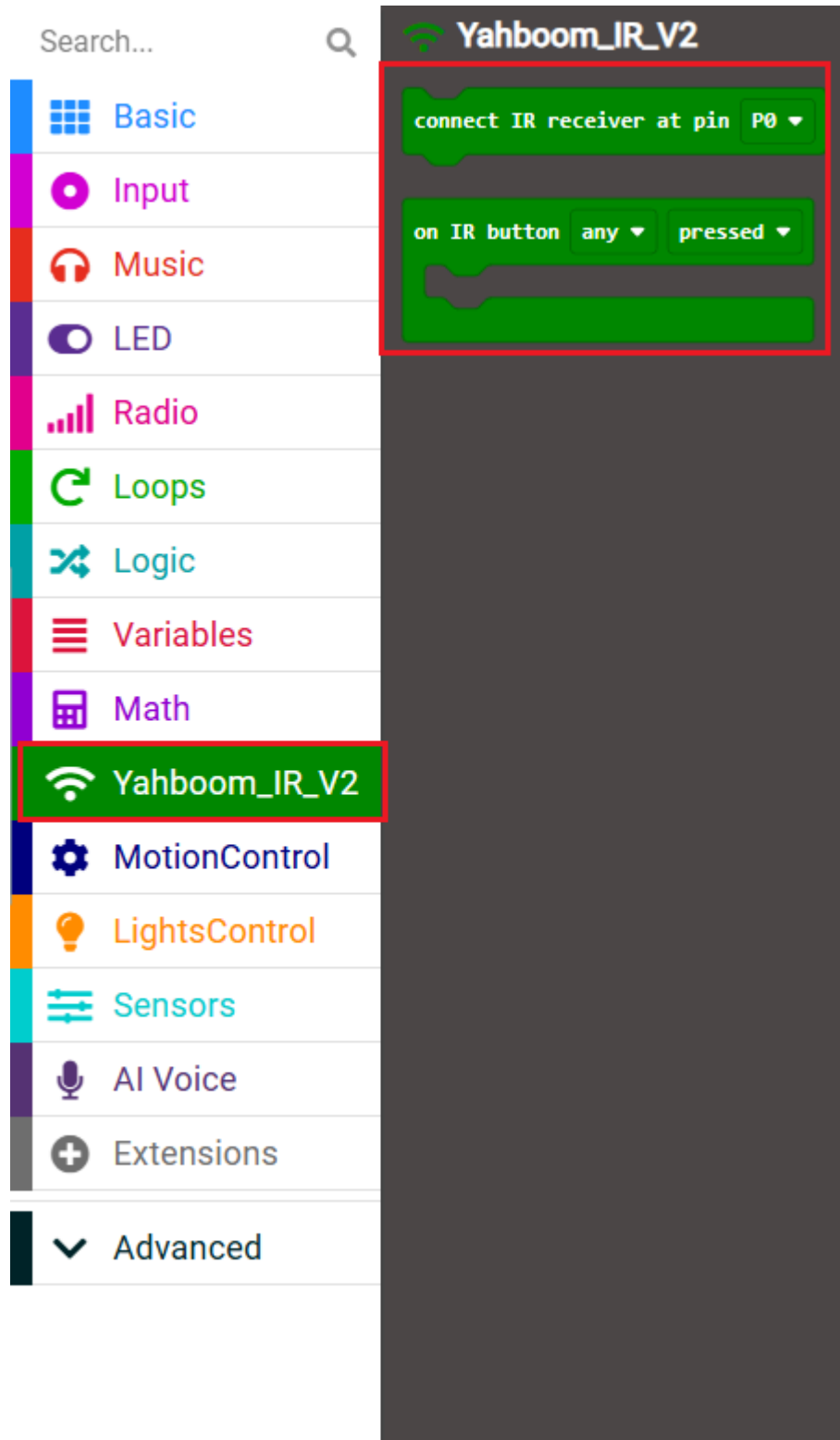


3. Drag `7.Infrared_Remote_Signal_Collection.hex` to the new project page. MakeCode will automatically import and open the project.

3. Experiment Source Code

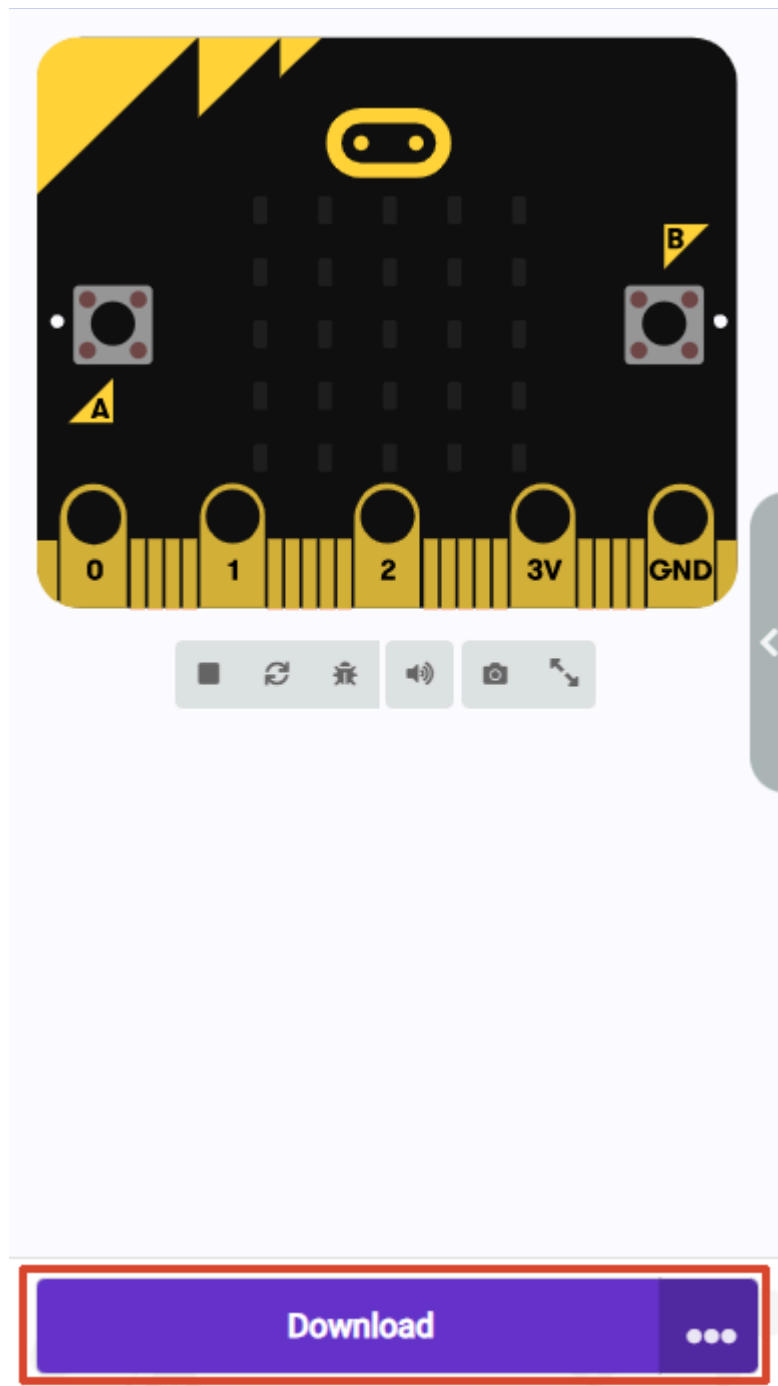


The building blocks used in this lesson are mainly located in the `Yahboom_IR_V2` category.



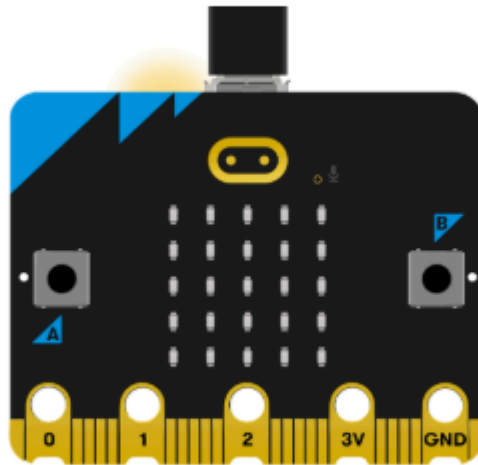
4. Program Download

1. Click the download button at the bottom left to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please connect the data cable to the Microbit first. If already connected, click [Next] here.

1. Connect your micro:bit to your computer



Next

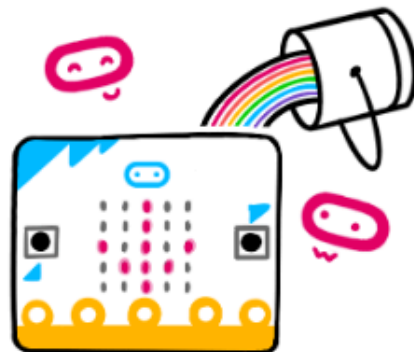
3. If the Microbit is detected as properly connected, the following interface will be displayed. At this point, you can click [Download].



Connected to micro:bit



Your micro:bit is connected! Pressing 'Download' will now automatically copy your code to your micro:bit.



Download

If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. You can choose between two methods: [Download as File] and [Pair].

2. Pair your micro:bit to your browser



Press the Pair button below.

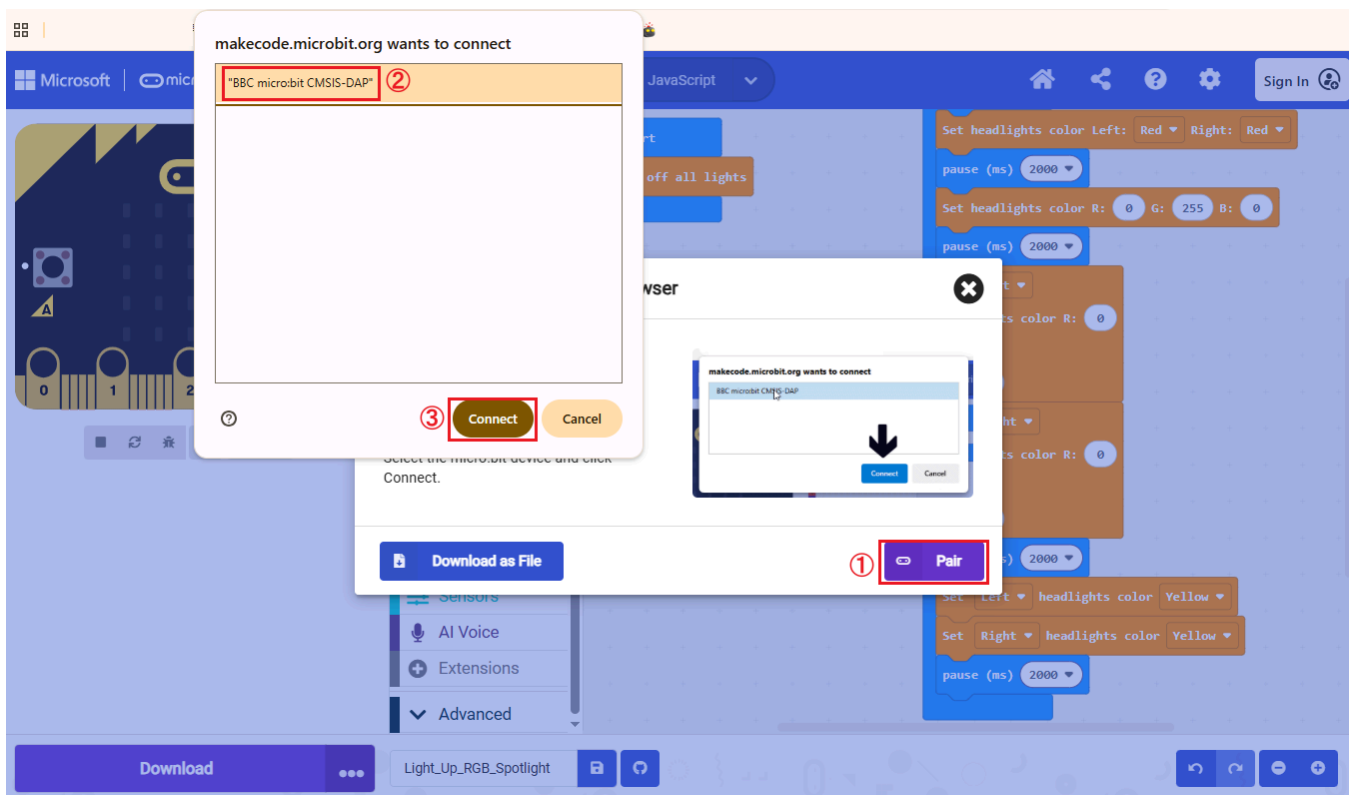
A window will appear in the top of your browser.

Select the micro:bit device and click Connect.

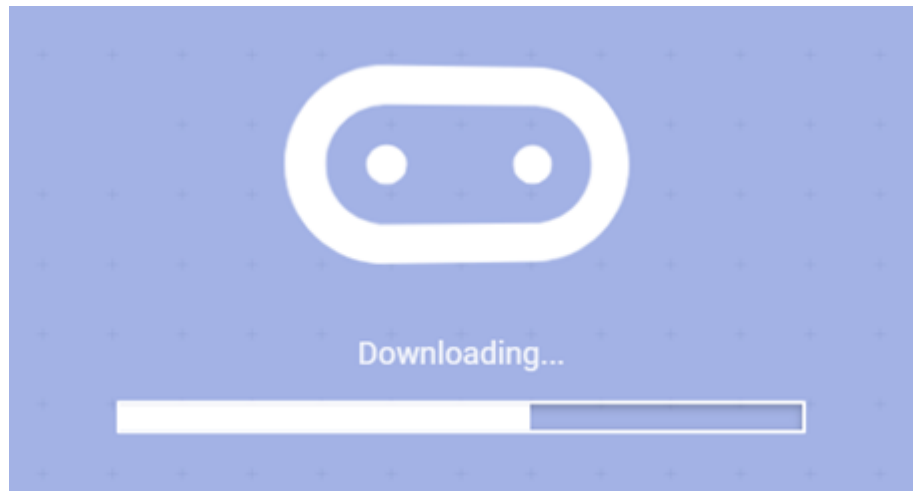


When selecting [Download as File], the system will download the currently opened program as `microbit-project-name.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive letter to complete the download.

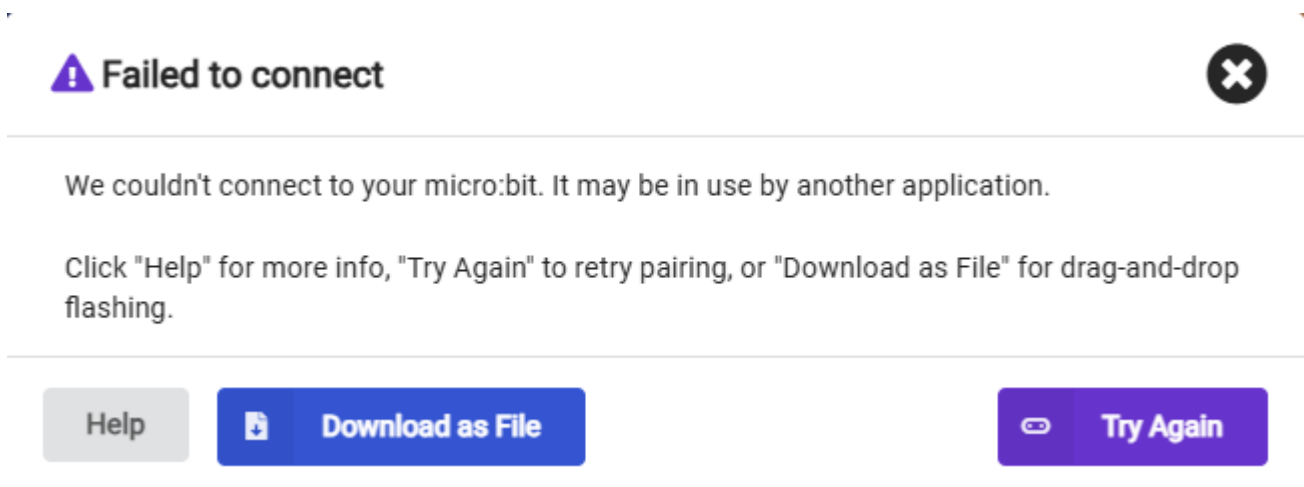
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC micro:bit CMSIS:DAP`.



4. When the following interface appears, it means the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



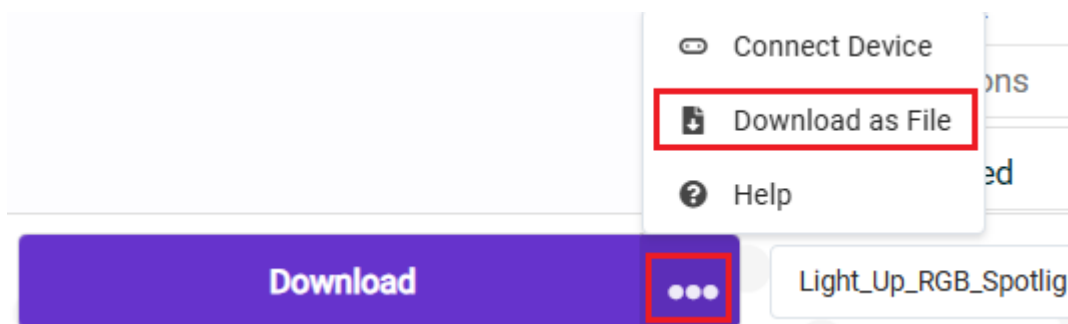
If multiple attempts continue to fail, please directly select [Download as File], save the current program as a file, and directly copy or drag the program to the corresponding MICROBIT drive letter to complete the download.



If you want to save the file directly, we provide two methods:

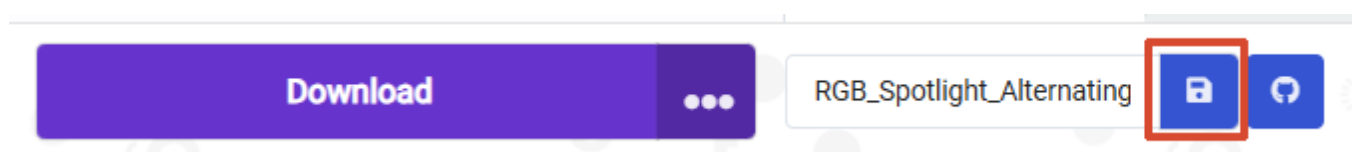
Method 1:

Click the [...] next to download, then select [Download as File]



Method 2:

Click [Save] after the [Project Name Edit Field].



5. Experiment Results

After flashing, turn on the switch (flip the switch to the ON position). The Micro:bit LED matrix will display a checkmark. At this point, you can use the remote control to operate.

The Bluetooth remote control has a plastic film at the bottom from the factory, which needs to be removed before use. It is recommended to control facing the infrared receiver probe. When remote controlling, it is recommended to press briefly. Wait for a period of time after each button press before pressing again.

User code: 00FF



Infrared Remote Button (Key Value)	Button Function
0x00	Display "Power"
0x80	Display "↑"
0x40	Display "Light"
0x20	Display "←"
0xA0	Display "Beep"
0x60	Display "→"

Infrared Remote Button (Key Value)	Button Function
0x10	Display "↩"
0x90	Display "↓"
0x50	Display "↗"
0x30	Increase the number on screen by 1 each press
0x70	Decrease the number on screen by 1 each press
0xB0	Display number 0
0x08	Display number 1
0x88	Display number 2
0x48	Display number 3
0x28	Display number 4
0xA8	Display number 5
0x68	Display number 6
0x18	Display number 7
0x98	Display number 8
0x58	Display number 9
0xFF	Controller user code (used only for identifying the remote control, no function)